Microbial Changes Upon Chemotherapy Induced GI Toxicity in GI Cancer Patients–a Population Study

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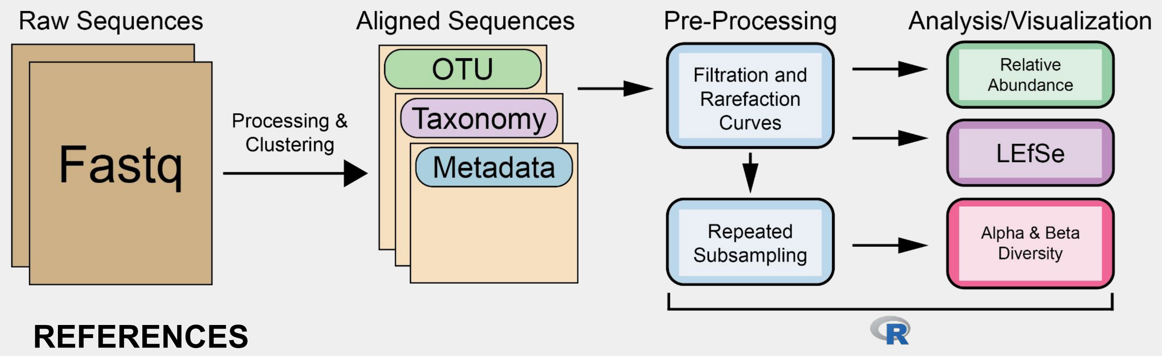
BACKGROUND

- Microbial drug metabolism can modulate chemotherapy efficacy and toxicity, linking microbiome composition to treatment outcomes. [1,2]
- Irinotecan, a broad-spectrum anticancer agent, disrupts the gut microbiome, where bacterial β-glucuronidase reactivates its metabolites (SN38G → SN38), driving diarrhea and mucositis. [3]
- Mild to severe GI toxicity occurs in 50–80% of patients treated with irinotecan, often leading to dose delays, reductions, or hospitalization. [4]

STUDY OVERVIEW

- **Approach:** We harmonize existing clinical microbiome datasets to compare microbiome changes upon treatment with irinotecan and other chemotherapy regimens.
- **Clinical data:** GI cancer patients grouped by condition (e.g., healthy, chemotherapy).
- **Goal:** Identify microbial signatures consistently associated with chemotherapy-induced GI toxicity from clinical data to guide mechanistic and lab-based translational studies.

METHODS



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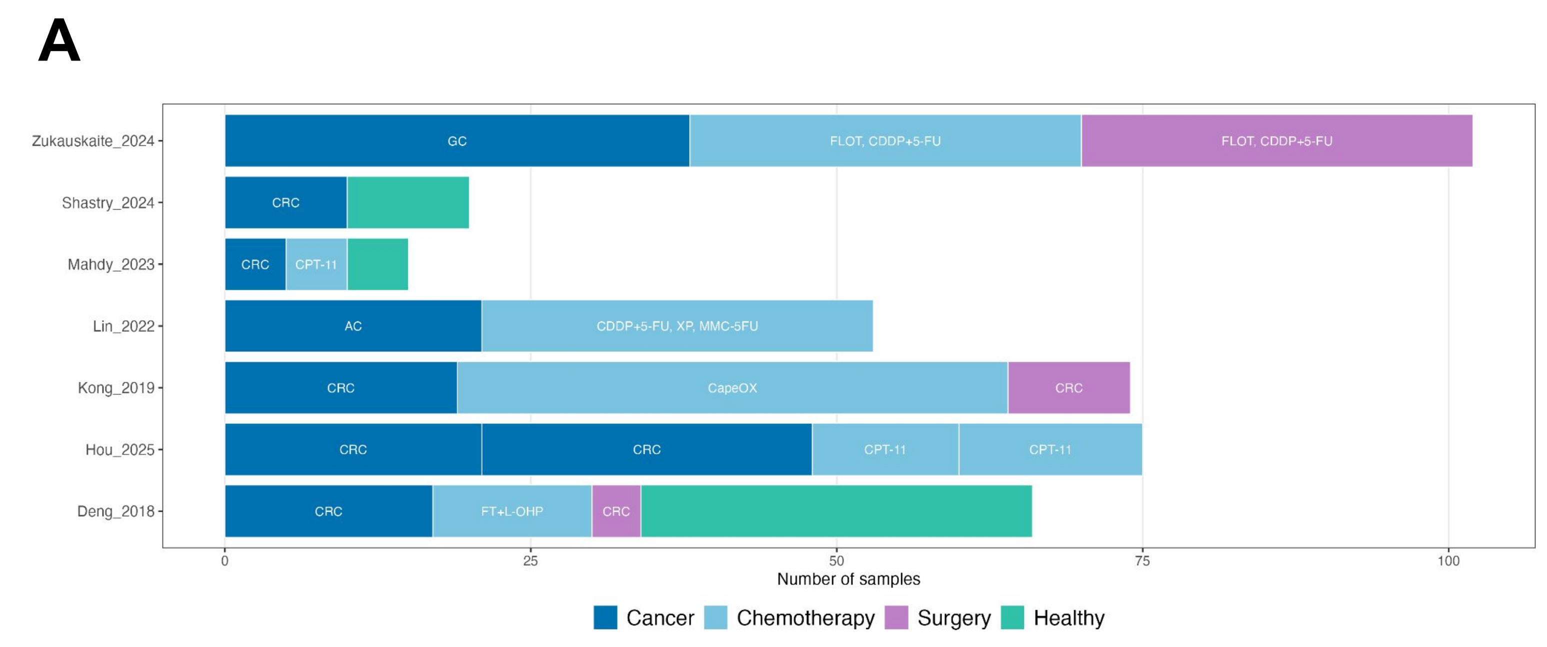
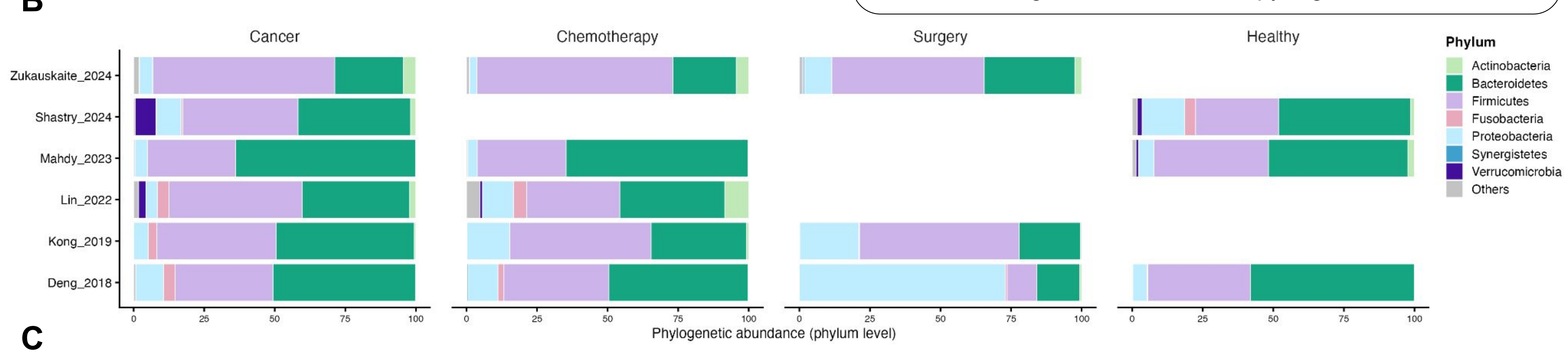
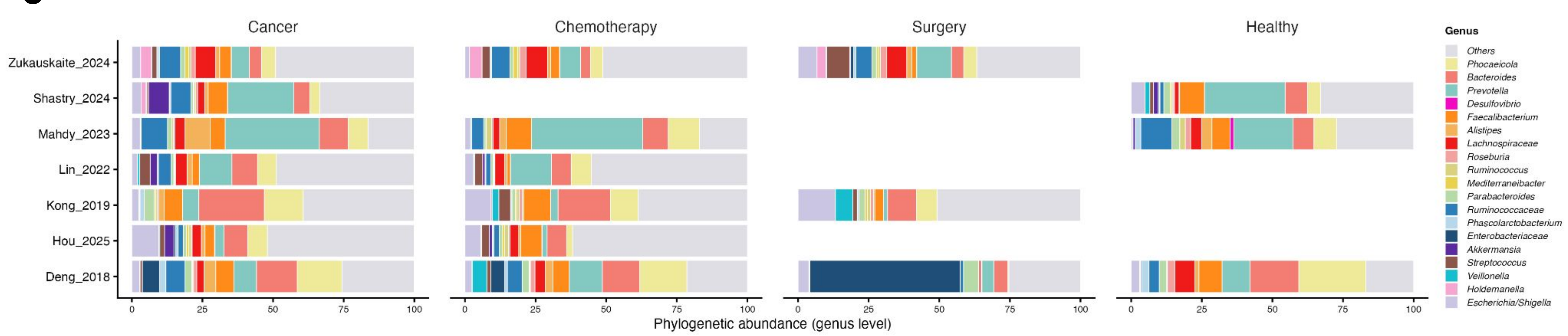
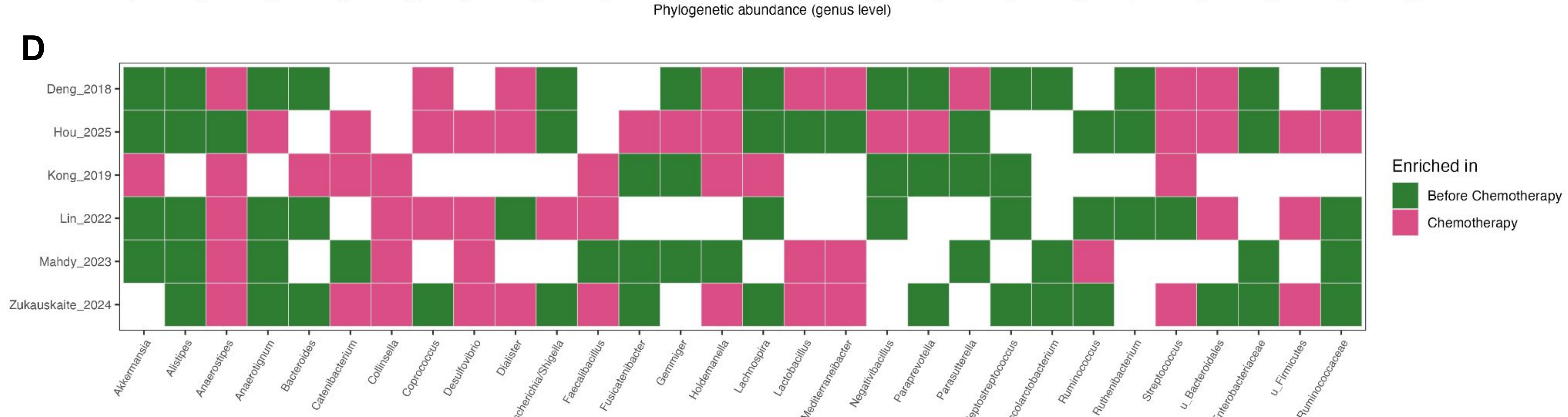
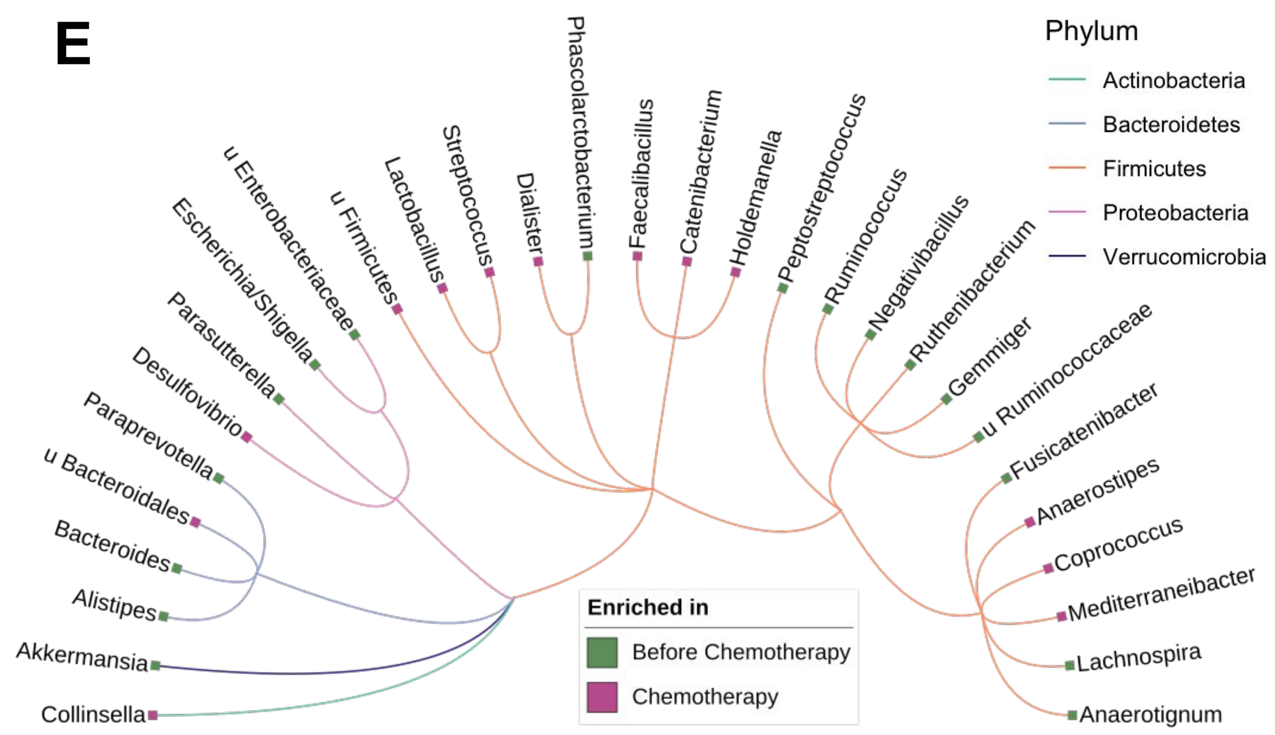


Chemotherapy is associated with distinct shifts in gut microbiome composition across multiple taxonomic levels. (A) Individuals per group across studies. **(B–C)** Taxonomic composition of the gut microbiota at the phylum **(B)** and genus **(C)** levels by group. **(D)** Heatmap of LEfSe differential abundance results between prechemotherapy and chemotherapy groups per study. Binary encoding indicates the group in which each genus was significantly enriched. Identified genera with <50% prevalence or <75% directional agreement across studies were excluded. **(E)** Phylogenetic tree of genera identified in **(D)**, colored by phylum, with tip labels indicating the enriched group. CRC, GC, AC: colon, gastric, anal cancer. Light blue: chemotherapy regimens.









RESULTS

- We observe microbial shifts associated with chemotherapy-induced GI toxicity in GI cancer studies.
- Desulfovibrio (DSV), a sulfate-reducing, β-glucuronidase–producing bacterium, emerged as a taxon consistently associated with the chemotherapy treatment group across human studies.
- These findings suggest that hydrogen sulfide produced by DSV may be a significant proinflammatory molecule for chemotherapy-induced GI toxicity.
- Collinsella, a gut bacterium associated with epithelial barrier dysfunction and proinflammatory signaling, was also enriched in chemotherapy-treated groups.

NEXT STEPS

- Apply our harmonized pipeline to more studies to build a comprehensive chemotherapy–microbiome map.

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